Example 1: If the random variables X, Y, Z have the means $\mu_x = 3, \mu_y = 5, \mu_z = 2$, the variances $\sigma_x^2 = 8$, $\sigma_y^2 = 12$, $\sigma_z^2 = 18$ and covariances $Cov(X, Y) = 1$, $Cov(X, Z) = -3$, $Cov(Y, Z) = 2$

If $U = X - 4Y + 2Z$ and $V = 3X - Y - Z$ functions are the linear combination of these three random variables. a) Find E(U), b) Find E(V), c) Find Var(U) d) Find Var(V) e) Find Cov(U,V),

If the random variables X, Y and Z are independent, find all the values again.

Solution:

$$U = X - 4Y + 2Z,\; V = 3X - Y - Z \quad \Rightarrow \quad X_1 = X, X_2 = Y, X_3 = Z; a_1 = 1,\; a_2 = -4,\; a_3 = 2\; ;\; b_1 = 3,\; b_2 = -1,\; b_3 = -1$$

$$U = a_1X_1 + a_2X_2 + a_3X_3,\; V = b_1X_1 + b_2X_2 + b_3X_3$$

$$E(U) = E(a_1X_1 + a_2X_2 + a_3X_3) = a_1E(X_1) + a_2E(X_2) + a_3E(X_3) = 1.3 + (-4).5 + 2.2 = -13$$

$$E(V) = E(b_1X_1 + b_2X_2 + b_3X_3) = b_1E(X_1) + b_2E(X_2) + b_3E(X_3) = 3.3 + (-1).5 + (-1).2 = 2$$

$$Var(U) = Var(a_1X_1 + a_2X_2 + a_3X_3) = a_1^2Var(X_1) + a_2^2Var(X_2) + a_3^2Var(X_3) + 2a_1a_2Cov(X_1, X_2) + 2a_1a_3Cov(X_1, X_3) + 2a_2a_3Cov(X_2, X_3)$$

$$= 1^2.8 + (-4)^2.12 + 2^2.18 + 2.1.(-4).1 + 2.1.2.(-3) + 2.(-4).2.2 = 8 + 192 + 72 - 8 - 12 - 32 = 220$$

$$Var(V) = Var(b_1X_1 + b_2X_2 + b_3X_3) = b_1^2Var(X_1) + b_2^2Var(X_2) + b_3^2Var(X_3) + 2b_1b_2Cov(X_1, X_2) + 2b_1b_3Cov(X_1, X_3) + 2b_2b_3Cov(X_2, X_3)$$

$$= 3^2.8 + (-1)^2.12 + (-1)^2.18 + 2.3.(-1).1 + 2.3.(-1).(-3) + 2.(-1).(-1).2 = 72 + 12 + 18 - 6 + 18 + 4 = 192$$

$$Y_1 = \sum_{i=1}^{n} a_i X_i \quad \text{and} \quad Y_2 = \sum_{i=1}^{n} b_i X_i \Rightarrow \boxed{\operatorname{cov}(Y_1, Y_2) = \sum_{i=1}^{n} a_i b_i \cdot \operatorname{var}(X_i) + \sum_{i<j}\sum (a_i b_j + a_j b_i) \cdot \operatorname{cov}(X_i, X_j)}$$

$$Cov(U,V) = a_1b_1Var(X_1) + a_2b_2Var(X_2) + a_3b_3Var(X_3) + (a_1b_2 + a_2b_1)Cov(X_1, X_2) + (a_1b_3 + a_3b_1)Cov(X_1, X_3) + (a_2b_3 + a_3b_2)Cov(X_2, X_3)$$

$$= 1.3.8 + (-4).(-1).12 + 2.(-1).18 + \big(1.(-1) + (-4).3\big).1 + \big(1.(-1) + 2.3\big).(-3) + \big((-4).(-1) + 2.(-1)\big).2$$

$$= 24 + 48 - 36 - 13 - 15 + 4 = 12$$

If X_1, X_2, X_3 random var*iables are independent then* cov*ariances will be zero;*

$$E(U) = E(a_1X_1 + a_2X_2 + a_3X_3) = a_1E(X_1) + a_2E(X_2) + a_3E(X_3) = 1.3 + (-4).5 + 2.2 = -13$$

$$E(V) = E(b_1X_1 + b_2X_2 + b_3X_3) = b_1E(X_1) + b_2E(X_2) + b_3E(X_3) = 3.3 + (-1).5 + (-1).2 = 2$$

$$Var(U) = Var(a_1X_1 + a_2X_2 + a_3X_3) = a_1^2Var(X_1) + a_2^2Var(X_2) + a_3^2Var(X_3)$$

$$= 1^2.8 + (-4)^2.12 + 2^2.18 = 8 + 192 + 72 = 272$$

$$Var(V) = Var(b_1X_1 + b_2X_2 + b_3X_3) = b_1^2Var(X_1) + b_2^2Var(X_2) + b_3^2Var(X_3)$$

$$= 3^2.8 + (-1)^2.12 + (-1)^2.18 = 72 + 12 + 18 = 102$$

$$Cov(U,V) = a_1b_1Var(X_1) + a_2b_2Var(X_2) + a_3b_3Var(X_3)$$

$$= 1.3.8 + (-4).(-1).12 + 2.(-1).18$$

$$= 24 + 48 - 36 = 36$$

Example 2: A joint probability function is given below.

$$f(x,y)=\begin{cases} \dfrac{xy^2}{12} & 0<x<c \qquad 0<y<2 \\ 0 & \text{elsewhere} \end{cases}$$

a) Find c?

b) Find joint distribution function(joint cumulative distribution).

c) Find marginal density of X and Y.

d) Find P(X>2, Y<1).

e) Find P[(X+Y)<4].

f) Find P[X>2|(Y>1)].

Solution:

a) $$\int_0^c\int_0^2 \frac{xy^2}{12}dydx = \frac{1}{12}\int_0^c \frac{xy^3}{3}\Big|_0^2 dx = \frac{2}{9}\int_0^c xdx = \frac{1}{9}c^2 = 1 \Rightarrow c = 3$$

b) $$F(x,y) = \begin{cases} 0 & x<0,\ y<0 \\ F_1(x,y) & 0\le x<3,\ 0\le y<2 \\ F_2(y) & x\ge 3,\ 0\le y<2 \\ F_3(x) & 0\le x<3,\ y\ge 2 \\ 1 & x>3,\ y>2 \end{cases}$$

$$F_1(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} \frac{xy^2}{12}dydx = \int_0^x\int_0^y \frac{xy^2}{12}dydx = \frac{x^2y^3}{72}$$

$$F_2(y) = \int_{-\infty}^{x}\int_{-\infty}^{y} \frac{xy^2}{12}dydx = \int_0^3\int_0^y \frac{xy^2}{12}dydx = \frac{y^3}{8}$$

$$F_3(x) = \int_{-\infty}^{x}\int_{-\infty}^{y} \frac{xy^2}{12}dydx = \int_0^x\int_0^2 \frac{xy^2}{12}dydx = \frac{x^2}{9}$$

$$F(x,y) = \begin{cases} 0 & x<0,\ y<0 \\ \dfrac{x^2y^3}{72} & 0\le x<3,\ 0\le y<2 \\ \dfrac{y^3}{8} & x\ge 3,\ 0\le y<2 \\ \dfrac{x^2}{9} & 0\le x<3,\ y\ge 2 \\ 1 & x>3,\ y>2 \end{cases}$$

c) $$g(x)=\int_{-\infty}^{\infty}\frac{xy^2}{12}dy=\int_0^2\frac{xy^2}{12}dy=\frac{2x}{9}\Rightarrow g(x)=\begin{cases}\frac{2x}{9}, & 0<x<3\\ 0, & \text{elsewhere}\end{cases}$$

$$h(y)=\int_{-\infty}^{\infty}\frac{xy^2}{12}dx=\int_0^3\frac{xy^2}{12}dx=\frac{3y^2}{8}\Rightarrow h(y)=\begin{cases}\frac{3y^2}{8}, & 0<y<2\\ 0, & \text{elsewhere}\end{cases}$$

d) $$P(X>2,Y<1)=\int_2^3\int_0^1\frac{xy^2}{12}dydx=\frac{5}{72}$$

e) P[(X+Y)<4]=?

$$P\left[(X+Y)<4\right]=\int_0^2\int_0^{4-y}\frac{xy^2}{12}dxdy=\int_0^2\frac{x^2y^2}{24}\Big|_0^{4-y}dy=\int_0^2\frac{y^2(4-y)^2}{24}dy=\int_0^2\frac{y^2\left(16-8y+y^2\right)}{24}dy=\frac{1}{24}\left[\frac{16y^3}{3}-2y^4+\frac{y^5}{5}\right]_0^2$$

$$=\frac{17.06}{24}=0.711$$

f) P[X>2|(Y>1)]=?

$$P\left[X>2\,|\,(Y>1)\right]=\frac{P(X>2,Y>1)}{P(Y>1)}=\frac{\int_2^3\int_1^2 \frac{xy^2}{12}dxdy}{\int_1^2 \frac{3y^2}{8}dy}=\frac{\frac{1}{12}\int_2^3 \frac{xy^3}{3}\Big|_1^2 dx}{\frac{y^3}{8}\Big|_1^2}=\frac{\frac{1}{12}\int_2^3 \frac{7x}{3}dx}{\frac{7}{8}}=\frac{\frac{1}{12}\left[\frac{7x^2}{6}\right]_2^3}{\frac{7}{8}}=\frac{\frac{35}{72}}{\frac{7}{8}}=\frac{35}{72}\cdot\frac{8}{7}=0.55\ .$$

Example 3: Joint probability density function of the random variables X and Y is given below:

$$f(x,y)=\begin{cases}\frac{3}{2}xy^2 & 0<x<2 \qquad 0<y<1\\ 0 & \text{elsewhere}\end{cases}$$

are X and Y independent?

Solution:

$$g(x)=\int_0^1 \frac{3}{2}xy^2dy=\frac{3}{2}\left[\frac{xy^3}{3}\right]_0^1=\frac{x}{2} \qquad h(y)=\int_0^2 \frac{3}{2}xy^2dx=\frac{3}{2}\left[\frac{x^2y^2}{2}\right]_0^2=3y^2$$

$$g(x)=\frac{x}{2} \qquad 0<x<2 \qquad\qquad h(y)=3y^2 \qquad 0<y<1$$

$$f(x,y)=g(x)\cdot h(y)\Rightarrow \frac{3}{2}xy^2=\frac{x}{2}\cdot 3y^2\Rightarrow \frac{3}{2}xy^2=\frac{3}{2}xy^2$$

X and Y are independent.

Example 4: The joint cumulative distribution function of the random variables X and Y is given below;

$$F(x,y)=\begin{cases}0 & x<0,\quad y<0\\ cx^2y^3 & 0\le x\le 2,\quad 0\le y\le 1\\ 1 & x>2,\quad y>1\end{cases}$$

a) Find c.

b) Find joint probability density function.

c) Find marginal density functions.

d) Are X and Y independent.

e) Find P(X<1|Y>0,5).

Solution: a)

$$F(x,y)=\begin{cases}0 & x<0,\quad y<0\\ cx^2y^3 & 0\le x\le 2,\quad 0\le y\le 1\\ 1 & x>2,\quad y>1\end{cases} \quad \Rightarrow F(0,0)=0,\ \ F(2,1)=1 \qquad \text{F(2,1)=c.}2^2.1=1 \qquad \Rightarrow c=\frac{1}{4}$$

b)

$$f(x,y)=\frac{\partial^2 F(x,y)}{\partial x \partial y} \quad \Rightarrow \quad f(x,y)=\begin{cases} \dfrac{3xy^2}{2} & 0 \le x \le 2, \ \ 0 \le y \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

c)

$$g(x)=\int_{-\infty}^{\infty} f(x,y)dy=\int_0^1 \frac{3xy^2}{2}dy=\frac{x}{2} \qquad h(y)=\int_{-\infty}^{\infty} f(x,y)dx=\int_0^2 \frac{3xy^2}{2}dx=3y^2$$

$$g(x)=\begin{cases} \dfrac{x}{2} & 0 \le x \le 2 \\ 0 & \text{elsewhere} \end{cases} \qquad h(y)=\begin{cases} 3y^2 & 0 \le y \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

d) $f(x,y)=g(x).h(y)=\dfrac{3xy^2}{2}$ so X and Y random variables are independent.

e)

$$P\left(X<1 \middle| Y>0,5\right)=\frac{P\left(X<1, Y>0,5\right)}{P\left(Y>0,5\right)}=\frac{\int_{0,5}^{1}\int_{0}^{1}\frac{3xy^2}{2}dxdy}{\int_{0,5}^{1}3y^2dy}=\frac{\frac{7}{32}}{\frac{7}{8}}=\frac{1}{4}$$

Example 5: The probability distribution function (cumulative distribution) of the vehicle numbers arriving at a fuel station in 10 minutes in a certain period of the day is given below. Find the probability function.

$$F(x)=\begin{cases} 0 & x<0 \\ 0,1 & 0\le x<1 \\ 0,25 & 1\le x<2 \\ 0,45 & 2\le x<3 \\ 0,75 & 3\le x<4 \\ 0,9 & 4\le x<5 \\ 1 & x\ge 5 \end{cases}$$

Solution: the probability function

$$f(x)=\begin{cases} 0 & x<0 \\ 0,1 & x=0 \\ 0,25-0,10=0,15 & x=1 \\ 0,45-0,25=0,2 & x=2 \\ 0,75-0,45=0,3 & x=3 \\ 0,90-0,75=0,15 & x=4 \\ 1.00-0,90=0,1 & x=5 \\ 0 & x>5 \end{cases}$$

Example 6: The probability density function f(x) is given below.

$$f(x)=\begin{cases} x+1 & -1<x<0 \\ 2x-6 & 3<x<c \\ 0 & \text{elsewhere} \end{cases}$$

a)Find c?

b)Find cumulative distribution F(x).

c)Find P(X≤c/3).

d)Find P(X>3,5).

Solution: a) $\int_{-\infty}^{\infty} f(x)dx=\int_{-1}^{0}(x+1)dx+\int_{3}^{c}(2x-6)dx=\left(\frac{x^2}{2}+x\right)_{-1}^{0}+\left(x^2-6x\right)_{3}^{c}=-\frac{1}{2}+1+c^2-6c-9+18=1$

$$c^2-6c-9+\frac{17}{2}=0 \qquad c=3\pm\sqrt{9-\frac{17}{2}}=3\pm\frac{\sqrt{2}}{2} \qquad \Rightarrow c=3+\frac{\sqrt{2}}{2}$$

b)

$$F(x)=\begin{cases} 0 & x<-1 \\ F_1(x) & -1\le x<0 \\ F_2(x) & 0\le x<3 \\ F_3(x) & 3\le x<3+\sqrt{2}/2 \\ 1 & x>3+\sqrt{2}/2 \end{cases}$$

$$F_1(x)=\int_{-\infty}^{x} f(x)dx=\int_{-1}^{x}(x+1)dx=\left(\frac{x^2}{2}+x\right)_{-1}^{x}=\frac{x^2}{2}+x-\frac{1}{2}+1=\frac{x^2}{2}+x+\frac{1}{2};\quad F_2(x)=\int_{-\infty}^{x} f(x)dx=\int_{-1}^{0}(x+1)dx=\left(\frac{x^2}{2}+x\right)_{-1}^{0}=\frac{1}{2}$$

$$F_3(x)=\int_{-\infty}^{x} f(x)dx=\int_{-1}^{0}(x+1)dx+\int_{3}^{x}(2x-6)dx=\frac{1}{2}+\left(x^2-6x\right)_3^x=\frac{1}{2}+x^2-6x-9+18=x^2-6x+\frac{19}{2}$$

$$f(x)=\begin{cases} x+1 & -1<x<0 \\ 2x-6 & 3<x<c \\ 0 & \text{elsewhere} \end{cases}$$

$$F(x)=\begin{cases} 0 & x<-1 \\ \dfrac{x^2+2x+1}{2} & -1\le x<0 \\ \dfrac{1}{2} & 0\le x<3 \\ \dfrac{2x^2-12x+19}{2} & 3\le x<3+\sqrt{2}/2 \\ 1 & x>3+\sqrt{2}/2 \end{cases}$$

c) $P\left(X\le\frac{c}{3}\right)=P\left(X\le 1+\frac{\sqrt{2}}{6}\right)=F(1+\frac{\sqrt{2}}{6})=\frac{1}{2}$

d) $P\left(X>3{,}5\right)=P\left(X>\frac{7}{2}\right)=1-P\left(X\le\frac{7}{2}\right)=1-F\left(\frac{7}{2}\right)=1-\dfrac{2\left(\frac{7}{2}\right)^2-12\left(\frac{7}{2}\right)+19}{2}=\frac{1}{4}$

Example 7: A coin is tossed three times. Let; X shows the number of head, Y shows the number of tails for the first two tossing. Find the joint probability distribution of X and Y.

Solution:

S	TTT	TTH	THT	HTT	THH	HTH	HHT	HHH
X	0	1	1	1	2	2	2	3
Y	2	2	1	1	1	1	0	0

The joint probability distribution of X and Y;

X \ Y	0	1	2
0	0	0	1/8
1	0	2/8	1/8
2	1/8	2/8	0
3	1/8	0	0

$$P(X,Y) \geq 0$$

$$\sum P(X,Y) = 1$$

Example 8: The probability density function of X random variable is given below;

$$f(x) = \begin{cases} \frac{4}{3}(1-x^3) & 0 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

a) Find E(X),

b) Find Variance and standard deviation,

c) Find the second moment about the origin (0).

d) Find the third moment about the mean (μ_3)

e) Find the fourth moment about the mean (μ_4)

Solution:

a) $$E(X) = m_1 = \frac{4}{3}\int_0^1 x(1-x^3)\,dx = \frac{4}{3}\left(\frac{x^2}{2} - \frac{x^5}{5}\right)\Big|_o^1 = \frac{2}{5}$$

b) $$Var(x)=\int_0^1 (x-m_1)^2 f(x)\,dx=\frac{4}{3}\int_0^1\left(x-\frac{2}{5}\right)^2(1-x^3)\,dx=\frac{4}{3}\int_0^1\left(x^2-\frac{4}{5}x+\frac{4}{25}\right)(1-x^3)\,dx$$

$$=\frac{4}{3}\left(\frac{x^3}{3}-\frac{x^6}{6}-\frac{4}{5}.\frac{x^2}{2}+\frac{4}{5}.\frac{x^5}{5}+\frac{4}{25}.x-\frac{4}{25}.\frac{x^4}{4}\right)\Big|_0^1=0{,}06$$

Standard deviation

$$\sigma=\sqrt{Var(x)}\Rightarrow\sigma=\sqrt{0{,}06}\Rightarrow\sigma=0{,}245$$

c) The second moment about the origin

$$E(X^2)=m_2=\frac{4}{3}\int_0^1 x^2(1-x^3)dx=\frac{4}{3}\int(x^2-x^5)dx=\frac{4}{3}\left(\frac{x^3}{3}-\frac{x^6}{6}\right)\Big|_o^1=\frac{2}{9}$$

$$Var(X)=m_2-m_1^2=\frac{2}{9}-\left(\frac{2}{5}\right)^2=\frac{14}{225}=0{,}062$$

d) The third moment about the mean (μ_3)

$$E(X^3)=m_3=\frac{4}{3}\int_0^1 x^3(1-x^3)dx=\frac{4}{5}\left[\frac{x^4}{4}-\frac{x^7}{7}\right]\Big|_o^1=\frac{4}{28}=\frac{1}{7}$$

$$E(X^4)=m_4=\frac{4}{3}\int_0^1 x^4(1-x^3)dx=\frac{4}{3}\left(\frac{x^5}{5}-\frac{x^8}{8}\right)\Big|_o^1=\frac{1}{10}$$

Moments about the mean of the first three order

$$\mu_1=E[X-E(X)]=E(X-m_1)=E(X)-m_1=m_1-m_1=0;$$

$$\mu_2=E[X-E(X)]^2=E\left[(X-m_1)^2\right]=E(X^2)-2m_1E(X)+m_1^2=m_2-m_1^2;$$

$$\mu_3=E[X-E(X)]^3=E\left[(X-m_1)^3\right]=E(X^3)-3m_1E(X^2)+3m_1^2E(X)-m_1^3=m_3-3m_1m_2+2m_1^3;$$

$$\mu_4=E[X-E(X)]^4=E\left[(X-m_1)^4\right]=E(X^4)-4m_1E(X^3)+6m_1^2E(X^2)-4m_1^3E(X)+m_1^4=m_4-4m_3m_1+6m_1^2m_2-4m_1^3m_1+m_1^4$$

$$\mu_3=m_3-3m_1m_2+2m_1^3=\frac{1}{7}-3\left(\frac{2}{5}\right)\left(\frac{2}{9}\right)+2\left(\frac{2}{5}\right)^3=\frac{1}{7}-3\left(\frac{4}{25}\right)+2\left(\frac{8}{125}\right)=0{,}004$$

e) The fourth moment about the mean (μ_4)

$$\mu_4=m_4-4m_1m_3+6m_1^2m_2 - 3m_1^4$$

$$\mu_4=\frac{1}{10}-4\cdot\frac{1}{7}\cdot\frac{2}{5}+6\cdot(\frac{2}{5})^2\cdot\frac{2}{9}-3\cdot(\frac{2}{5})^4$$

$$\mu_4=0{,}00796$$

Example 9: The joint probability distribution of X and Y random variables is given below.

Find the covariance of X and Y,

		X		
		-1	0	1
Y	-1	1/6	1/3	1/6
	0	0	0	0
	1	1/6	0	1/6

Solution:

		X			
		-1	0	1	h(y)
Y	-1	1/6	1/3	1/6	2/3
	0	0	0	0	0
	1	1/6	0	1/6	1/3
g(x)		1/3	1/3	1/3	

$$E(X)=\mu_x=\sum_{x=-1}^{1}\sum_{y=-1}^{1}x.f(x,y)=\sum_{x=-1}^{1}x\left(\sum_{y=-1}^{1}f(x,y)\right)$$

$$=\sum_{x=-1}^{1}x.g(x)=(-1).\frac{1}{3}+0.\frac{1}{3}+1.\frac{1}{3}=0$$

$$E(Y)=\mu_y=\sum_{x=-1}^{1}\sum_{y=-1}^{1}y.f(x,y)=\sum_{y=-1}^{1}y\left(\sum_{x=-1}^{1}f(x,y)\right)$$

$$=\sum_{y=-1}^{1}y.h(y)=(-1).\frac{2}{3}+0.0+1.\frac{1}{3}=-\frac{1}{3}$$

$$E(XY)=\sum_{x=-1}^{1}\sum_{y=-1}^{1}x.y.f(x,y)=\sum_{x=-1}^{1}x\left(\sum_{y=-1}^{1}y.f(x,y)\right)$$

$$=\sum_{x=-1}^{1}x.\big((-1).f(x,-1)+0.f(x,0)+1.f(x,1)\big)$$

$$=(-1).\big((-1).f(-1,-1)+1.f(-1,1)\big)+1.\big((-1).f(1,-1)+1.f(1,1)\big)$$

$$\Rightarrow E(XY)=0$$

$$\sigma_{xy}=Cov(X,Y)=E(XY)-\mu_x\mu_y=0-0.\left(-\frac{1}{3}\right)=0$$

Example 10: The joint distribution function (joint cumulative distribution) of X and Y random variables is given below. Find the probability of $P(1 < X < 3, 1 < Y < 2)$.

$$F(x,y)=\begin{cases}\left(1-e^{-x}\right)\left(1-e^{-y}\right) & x>0 \quad y>0\\ 0 & \text{elsewhere}\end{cases}$$

$$P\left(1<X<3,1<Y<2\right)=\int_1^3\int_1^2 f(x,y)dydx; \quad f(x,y)=\frac{\partial^2 F(x,y)}{\partial x\partial y}=\begin{cases}e^{-(x+y)} & x>0 \quad y>0\\ 0 & \text{elsewhere}\end{cases}$$

$$P\left(1<X<3,1<Y<2\right)=\int_1^3\int_1^2 f(x,y)dydx=\int_1^3\int_1^2 e^{-(x+y)}dydx=e^{-5}-e^{-4}-e^{-3}+e^{-2}$$

Example 11: The joint probability distribution of X,Y and Z random variables is given below.

$$f(x,y.z)=\begin{cases}k\left(x+y\right)z & x=1,2;\, y=1,2,3;\, z=1,2\\ 0 & \text{elsewhere}\end{cases}$$

a) Find the value of k,
b) Find P(X=2, Y+Z≤3),
c) Find the joint marginal distribution of X and Y,
d) Find the conditional distribution of Z given X=1, Y=2,
e) Find the marginal distributions of X, Y and Z.,are these random variables independent, find the means of all random variables.
f) Find the joint distribution of X and Y given Z=1.

a) $i)\ \text{For } \forall (x,y,z) \quad f(x,y,z) \ge 0 \Rightarrow k \ge 0$ $\qquad ii)\ \sum_x \sum_y \sum_z f(x,y,z) = 1$

$$\sum_{x=1}^{2}\sum_{y=1}^{3}\sum_{z=1}^{2} f(x,y,z) = \sum_{x=1}^{2}\sum_{y=1}^{3}\left(f(x,y,1)+f(x,y,2)\right) = \sum_{x=1}^{2}\left(f(x,1,1)+f(x,1,2)\right)+\left(f(x,2,1)+f(x,2,2)\right)+\left(f(x,3,1)+f(x,3,2)\right)$$

$$=\left(f(1,1,1)+f(1,1,2)\right)+\left(f(1,2,1)+f(1,2,2)\right)+\left(f(1,3,1)+f(1,3,2)\right)+\left(f(2,1,1)+f(2,1,2)\right)$$

$$+\left(f(2,2,1)+f(2,2,2)\right)+\left(f(2,3,1)+f(2,3,2)\right)$$

$$= 2k+4k+3k+6k+4k+8k+3k+6k+4k+8k+5k+10k = 63k = 1 \quad \Rightarrow \quad k = \frac{1}{63}$$

$$f(x,y,z) = \begin{cases} \dfrac{(x+y)z}{63} & x=1,2;\ y=1,2,3;\ z=1,2 \\ 0 & \text{elsewhere} \end{cases}$$

b) $P(X=2, Y+Z \le 3) = f(2,1,1)+f(2,1,2)+f(2,2,1) = \frac{3}{63}+\frac{6}{63}+\frac{4}{63} = \frac{13}{63}$

c) $g_1(x,y) = \sum_{z=1}^{2}\frac{(x+y)z}{63} = \frac{(x+y)}{63}+\frac{(x+y)2}{63} = \frac{(x+y)}{21},$

$$g_1(x,y) = \begin{cases} \dfrac{(x+y)}{21} & x=1,2;\ y=1,2,3 \\ 0 & \text{elsewhere} \end{cases}$$

		Y		
		1	2	3
X	1	2/21	3/21	4/21
	2	3/21	4/21	5/21

d) $$f(z|x=1,y=2)=\frac{f(1,2,z)}{g_1(1,2)}=\frac{f(1,2,z)}{3/21}=\frac{21}{3}\frac{(1+2)z}{63}=\frac{z}{3}\quad\Rightarrow f(z|x=1,y=2)=\begin{cases}\frac{z}{3} & z=1,2\\ 0 & \text{elsewhere}\end{cases}$$

Z	1	2
$f(z\|x=1,y=2)$	1/3	2/3

e)
$$g_2(x)=\sum_{y=1}^{3}\sum_{z=1}^{2}\frac{(x+y)z}{63}=\sum_{y=1}^{3}\left(\frac{(x+y)}{63}+\frac{(x+y)2}{63}\right)=\sum_{y=1}^{3}\frac{(x+y)}{21}=\frac{(x+1)}{21}+\frac{(x+2)}{21}+\frac{(x+3)}{21}=\frac{x+2}{7},\qquad g_2(x)=\begin{cases}\frac{x+2}{7} & x=1,2\\ 0 & \text{elsewhere}\end{cases}$$

$$g_3(y)=\sum_{x=1}^{2}\sum_{z=1}^{2}\frac{(x+y)z}{63}=\sum_{x=1}^{2}\left(\frac{(x+y)}{63}+\frac{(x+y)2}{63}\right)=\sum_{x=1}^{2}\frac{(x+y)}{21}=\frac{(1+y)}{21}+\frac{(2+y)}{21}=\frac{3+2y}{21},\qquad g_3(y)=\begin{cases}\frac{3+2y}{21} & y=1,2,3\\ 0 & \text{elsewhere}\end{cases}$$

$$g_4(z)=\sum_{x=1}^{2}\sum_{y=1}^{3}\frac{(x+y)z}{63}=\sum_{x=1}^{2}\left(\frac{(x+1)z}{63}+\frac{(x+2)z}{63}+\frac{(x+3)z}{63}\right)=\sum_{x=1}^{2}\frac{(xz+2z)}{21}=\frac{(z+2z)}{21}+\frac{(2z+2z)}{21}=\frac{z}{3},\; g_4(z)=\begin{cases}\frac{z}{3} & z=1,2\\ 0 & \text{elsewhere}\end{cases}$$

X	1	2
$g_2(x)$	3/7	4/7

Y	1	2	3
$g_3(y)$	5/21	7/21	9/21

Z	1	2
$g_4(z)$	1/3	2/3

$$f(1,1,1)=\frac{2}{63}\neq g_2(1).g_3(1).g_4(1)\qquad E(X)=\mu_x=\frac{11}{7}\qquad E(Y)=\mu_y=\frac{46}{21}\qquad E(Z)=\mu_z=\frac{5}{3}$$

f)

$$f(x,y|z=1)=\frac{f(x,y,1)}{g_4(1)}=\frac{f(x,y,1)}{1/3}=3\frac{(x+y)}{63}=\frac{x+y}{21} \qquad \Rightarrow f(x,y|z=1)=\begin{cases}\frac{x+y}{21} & x=1,2;\,y=1,2,3\\ 0 & \text{elsewhere}\end{cases}$$

		Y		
		1	2	3
X	1	2/21	3/21	4/21
	2	3/21	4/21	5/21

Example 12: The joint probability density function of X_1, X_2, X_3 random variables is given below.

$$f(x_1,x_2.x_3)=\begin{cases}(x_1+x_2)e^{-x_3} & 0<x_1<1, 0<x_2<1, x_3>0\\ 0 & \text{elsewhere}\end{cases}$$

a) Find the joint marginal density of X_1, X_3 ,
b) Find the marginal density of X_1 ,
c) Find Var(X_1),
d) Find the marginal density of X_2 and X_3 ,are these random variables independent?,
e) Find the conditional density of X_2 given X_1=0,5 and X_3 =1.

Solution: a)

$$g_1(x_1,x_3)=\int_{-\infty}^{\infty} f(x_1,x_2.x_3)dx_2=\int_0^1 (x_1+x_2)e^{-x_3}dx_2=\left(x_1+\frac{1}{2}\right)e^{-x_3}; \qquad \Rightarrow g_1(x_1,x_3)=\begin{cases}\left(x_1+\frac{1}{2}\right)e^{-x_3} & 0<x_1<1,\ x_3>0\\ 0 & \text{elsewhere}\end{cases}$$

b) $$g_2(x_1)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x_1,x_2.x_3)dx_2dx_3=\int_0^{\infty}\int_0^1 (x_1+x_2)e^{-x_3}dx_2dx_3=\lim_{b\to\infty}\int_0^b\left(x_1+\frac{1}{2}\right)e^{-x_3}dx_3=\lim_{b\to\infty}\left(x_1+\frac{1}{2}\right)\left(-e^{-x_3}\right)_0^b=x_1+\frac{1}{2}$$

$$\Rightarrow g_2(x_1)=\begin{cases}x_1+\frac{1}{2} & 0<x_1<1\\ 0 & \text{elsewhere}\end{cases}$$

c)

$$Var(X_1)=E(X_1^2)-\mu_{x_1}^{\ 2}; \qquad E(X_1)=\mu_{x_1}=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 f(x_1,x_2.x_3)dx_1dx_2dx_3=\int_{-\infty}^{\infty} x_1 g_2(x_1)dx_1=\int_0^1 x_1\left(x_1+\frac{1}{2}\right)dx_1=\frac{1}{3}+\frac{1}{4}=\frac{7}{12}$$

$$E(X_1^2)=\int_{-\infty}^{\infty} x_1^{\ 2} g_2(x_1)dx_1=\int_0^1 x_1^{\ 2}\left(x_1+\frac{1}{2}\right)dx_1=\frac{1}{4}+\frac{1}{6}=\frac{5}{12} \qquad \Rightarrow Var(X_1)=E(X_1^2)-\mu_{x_1}^{\ 2}=\frac{7}{12}-\left(\frac{5}{12}\right)^2=\frac{59}{144}$$

d) $$g_3(x_2)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x_1,x_2.x_3)dx_1dx_3=\int_0^{\infty}\int_0^1 (x_1+x_2)e^{-x_3}dx_1dx_3=\lim_{b\to\infty}\int_0^b\left(\frac{1}{2}+x_2\right)e^{-x_3}dx_3=\lim_{b\to\infty}\left(\frac{1}{2}+x_2\right)\left(-e^{-x_3}\right)_0^b=\frac{1}{2}+x_2$$

$$\Rightarrow g_3(x_2)=\begin{cases}\frac{1}{2}+x_2 & 0<x_2<1\\ 0 & \text{elsewhere}\end{cases}$$

$$g_4(x_3)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x_1,x_2.x_3)dx_1dx_2=\int_0^1\int_0^1(x_1+x_2)e^{-x_3}dx_1dx_2=\int_0^1\left(\frac{1}{2}+x_2\right)e^{-x_3}dx_2=e^{-x_3}$$

$$\Rightarrow g_4(x_3)=\begin{cases}e^{-x_3} & x_3>0\\ 0 & \text{elsewhere}\end{cases} \qquad \Rightarrow f(x_1,x_2.x_3)\neq g_2(x_1).g_3(x_2).g_4(x_3)$$

e) $$f(x_2|x_1=0,5;x_3=1)=\frac{f(0,5;x_2;1)}{g_1(0,5;1)}=\frac{(0,5+x_2)e^{-1}}{\left(0,5+\frac{1}{2}\right)e^{-1}}=(0,5+x_2)\Rightarrow f(x_2|x_1=0,5;x_3=1)=\begin{cases}0,5+x_2 & 0<x_2<1\\ 0 & \text{elsewhere}\end{cases}$$

Example 13: The probability density function of X_1, X_2, X_3 random variables are given below. Find the probability of $P(X_1+X_2\leq 1, X_3>1)$

$$f_1(x_1)=\begin{cases}e^{-x_1} & x_1>0\\ 0 & \text{elsewhere}\end{cases};\quad f_2(x_2)=\begin{cases}2e^{-2x_2} & x_2>0\\ 0 & \text{elsewhere}\end{cases};\quad f_3(x_3)=\begin{cases}3e^{-3x_3} & x_3>0\\ 0 & \text{elsewhere}\end{cases}$$

Solution: $$f(x_1,x_2,x_3)=f_1(x_1).f_2(x_2).f_3(x_3)=\begin{cases}6e^{-(x_1+2x_2+3x_3)} & x_1>0, x_2>0, x_3>0\\ 0 & \text{elsewhere}\end{cases}$$

$$P(X_1+X_2\leq 1, X_3>1)=\int_1^{\infty}\int_0^1\int_0^{1-x_2} f(x_1,x_2,x_3)dx_1dx_2dx_3=\int_1^{\infty}\int_0^1\int_0^{1-x_2} 6e^{-(x_1+2x_2+3x_3)}dx_1dx_2dx_3=0,020$$

Example 14:

Let X be a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Let also

$$Y = g(X) = \begin{cases} X & 0 \leq X \leq \frac{1}{2} \\ \frac{1}{2} & X > \frac{1}{2} \end{cases}$$

Find the CDF of Y.

Solution:

First we note that $R_X = [0, 1]$. For $x \in [0, 1]$, $0 \leq g(x) \leq \frac{1}{2}$. Thus, $R_Y = [0, \frac{1}{2}]$, and therefore

$$F_Y(y) = 0, \qquad \text{for } y < 0,$$

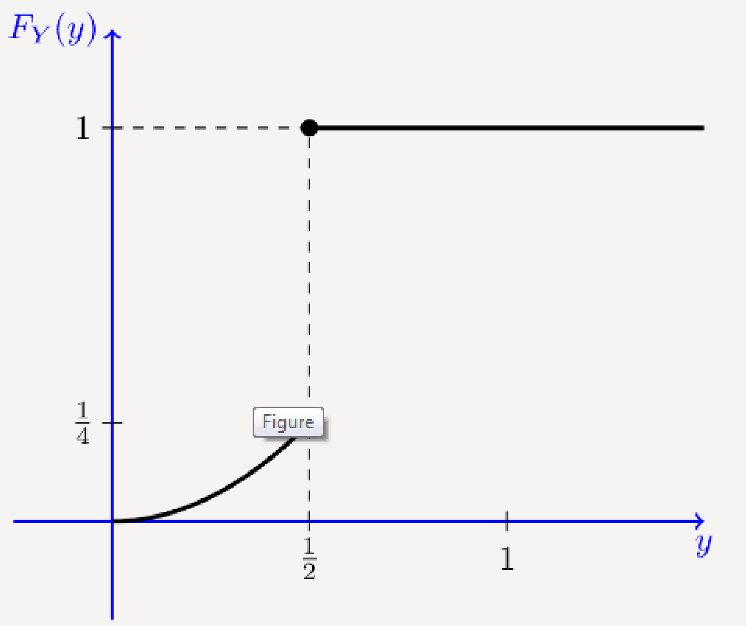
$$F_Y(y) = 1, \qquad \text{for } y > \frac{1}{2}.$$

Now note that

$$P\left(Y = \tfrac{1}{2}\right) = P\left(X > \tfrac{1}{2}\right)$$
$$= \int_{\frac{1}{2}}^{1} 2x dx = \tfrac{3}{4}.$$

Also, for $0 < y < \frac{1}{2}$,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X \leq y) \\ &= \int_0^y 2x dx \\ &= y^2. \end{aligned}$$



Thus, the CDF of Y is given by

$$F_Y(y) = \begin{cases} 1 & y \geq \frac{1}{2} \\ y^2 & 0 \leq y < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

Figure shows the CDF of Y. We note that the CDF is not continuous, so Y is not a continuous random variable. On the other hand, the CDF is not in the staircase form, so it is not a discrete random variable either. It is indeed a *mixed* random variable. There is a jump at $y = \frac{1}{2}$, and the amount of jump is $1 - \frac{1}{4} = \frac{3}{4}$, which is the probability that $Y = \frac{1}{2}$. The CDF is continuous at other points.

$$F_Y(y) = \begin{cases} 1 & y \geq \frac{1}{2} \\ y^2 & 0 \leq y < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

The CDF of Y has a continuous part and a discrete part. In particular, we can write

$$F_Y(y) = C(y) + D(y),$$

where $C(y)$ is the continuous part of $F_Y(y)$, i.e.,

$$C(y) = \begin{cases} \frac{1}{4} & y \geq \frac{1}{2} \\ y^2 & 0 \leq y < \frac{1}{2} \\ 0 & y < 0 \end{cases}$$

The discrete part of $F_Y(y)$ is $D(y)$, given by

$$D(y) = \begin{cases} \frac{3}{4} & y \geq \frac{1}{2} \\ 0 & y < \frac{1}{2} \end{cases}$$

Example 15: Let X be a continuous random variable with the following PDF:

$$f(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2 \\ 0, & elsewhere \end{cases}$$

We define $Y = g(X)$, where the function $g(x)$ is defined as

$$g(x) = \begin{cases} 1 & x > 1 \\ x & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the CDF of Y.

Solution:

Note that $R_Y = [0, 1]$. Therefore,

$$F_Y(y) = 0, \quad \text{for} \quad y < 0,$$

$$F_Y(y) = 1, \quad \text{for} \quad y \ge 1.$$

We also note that

$$P(Y = 0) = P(X < 0) = \frac{1}{2}, \qquad \int_{-2}^{0} \frac{1}{4} dx = \frac{1}{4} x \Big|_{-2}^{0} = \frac{1}{2}$$

$$P(Y = 1) = P(X > 1) = \frac{1}{4}. \qquad \int_{1}^{2} \frac{1}{4} dx = \frac{1}{4} x \Big|_{1}^{2} = \frac{1}{4}$$

Also for $0 < y < 1$,

$$F_Y(y) = P(Y \le y) = P(X \le y) = F_X(y) = \frac{y+2}{4}.$$

$$\int_{-2}^{y} \frac{1}{4}\,dx = \frac{1}{4}x\Big|_{-2}^{y} = \frac{1}{4}(y+2)$$

Thus, the CDF of Y is given by

$$F_Y(y) = \begin{cases} 1 & y \ge 1 \\ \frac{y+2}{4} & 0 \le y < 1 \\ 0 & \text{otherwise} \end{cases}$$

In particular, we note that there are two jumps in the CDF, one at $y = 0$ and another at $y = 1$.